Class 1: Difference-in-Differences

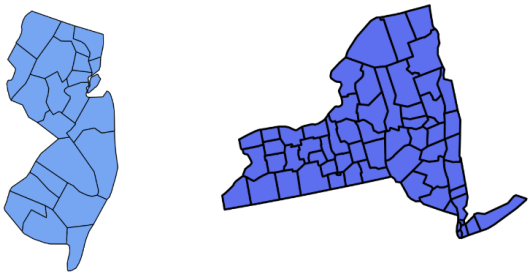
In-class student worksheet

Name: _____ Date: _____

Learning goals: construct a counterfactual, calculate and interpret DiD, and explain the assumption needed for a causal interpretation.

1. Why is a simple comparison insufficient?

Suppose State A has higher average UI benefits and longer average UI receipt than State B.



	NJ	NY
Avg. UI benefit ($)	628.29	553.97
Avg. UI duration (weeks)	19.3	17.9

Can we conclude that higher benefits caused longer UI receipt? Give **two other explanations** for the difference between the states.

2. A policy change: New Jersey and Pennsylvania

In 1996, New Jersey temporarily made 13 additional weeks of UI benefits available. Pennsylvania did not introduce the same extension. We compare regular-benefit exhaustion rates before and after the extension.

Outcome: the percentage of claimants who use up their *regular* UI benefits. It is not the unemployment rate and does not measure exhaustion of the additional extension.

A. Identify the design. On the axes below, plot both states' before/after outcomes. Add NJ's counterfactual after-period point. Label the estimated policy effect.



Treated state: _____ Comparison state: _____

What is the treatment? _____

What is the outcome of their interest? _____

B. Complete the table. All differences are in **percentage points**.

	Before	After	After − Before
New Jersey	49.9%	49.4%	
Pennsylvania	31.8%	29.0%	
NJ − PA			

C. Calculate the DiD estimate using both approaches.

$$\widehat{\tau} = (\text{NJ after} - \text{NJ before}) - (\text{PA after} - \text{PA before}) = _____$$

D. Interpret your estimate in one sentence.

3. When is the comparison credible?

1. Parallel trends: _____

    - How do we know there are parallel trends?

2. The policy changes is the only important change that happened in NJ is this time period.

    - Example: If NJ experienced a larger economic downturn than PA at the same time as the extension, would the DiD estimate be biased upward or downward?

4. DiD with Linear Regression:

Regression connection. In the equation below, circle the coefficient that corresponds to your DiD estimate:

$$Y_{i,t} = \alpha + \beta NJ_i + \lambda After_t + \tau(NJ_i \times After_t) + \varepsilon_{i,t}.$$

$NJ_i = 1$ for NJ and 0 for PA; $After_t = 1$ after the extension and 0 before.

Table 1: Predicted Regular UI Exhaustion Rates

	Before	After	After − Before
NJ			
PA			
NJ − PA			